Excel file with groups & researches

NIRS_assignment_groups.xlsx

Dataset

Cognitive tasks: 20s baseline+ 28s task+ 20s recovery

Motor task: 20s baseline+ 10s task+ 20s recovery

Necker cube: 20s baseline+ 20s task+ 20s recovery

Rotating snakes: 20s baseline+ 20s task+ 20s recovery

Flickering wedges: 20s baseline+ 10s task+ 20s recovery

TD NIRS data were performed using devices with 4096 channels of 3 ps per channel, and the files contain 5 repetition at 4 wavelengths: 670nm, 730nm, 780nm, 830nm

Research questions

- **TD-NIRS Data**:
  Compute the Differential Pathlength Factor (DPF) for each optode placement.
    - Do you observe any differences across different regions of the head, or in comparison with values reported in the literature?
- **CW-NIRS Data**:
    - Are all signals (i.e., all channels) of acceptable quality?
    - Do you notice any artifacts? If so, can they be removed and how?
    - Calculate the changes in oxy- and deoxy-hemoglobin (HbO and HbR) concentrations during the experiments.
    - Perform block averaging for each task and analyze whether there are any significant differences across tasks—e.g., comparing tasks of varying complexity.

Deliverable

One zipped file “Group_n_XX” including:

- Folder with script/scripts used for NIRS processing;

- Folder with processing output;

- Report of maximum 5 pages and 4 figures or tables including introduction, aim, materials and methods, results, and brief discussion & conclusions (see template for the Executive Summary at POLIMI website).

Deadline:

April 29th, 2026, Webeep folder “Consegne --> NIRS_assignment”.